Magnetically Aligned CNT Arrays for Scalable Gecko-Inspired Dry Adhesion

Ki-Hoon Lee, ME 218N, Department of Mechanical Engineering, UC Berkeley, May 2025

Abstract:

Geckos exhibit remarkable dry adhesion through arrays of microscopic setae, inspiring the development of synthetic adhesives that replicate this biological strategy. Traditional approaches, such as vertically aligned carbon nanotubes (CNTs) grown via chemical vapor deposition (CVD), achieve strong adhesion but suffer from high cost, substrate rigidity, and complex fabrication requirements. This project presents a low-cost, scalable alternative by magnetically aligning multi-walled carbon nanotubes (MWCNTs) embedded with magnetic nanoparticles within a flexible adhesive substrate. By applying a uniform magnetic field during curing, partial vertical alignment is achieved, enhancing surface contact and dry adhesion. Simulation of the alignment process, modeled via magnetic torque dynamics, showed increasing alignment sharpness with higher field strengths. Adhesion performance was estimated using Johnson-Kendall-Roberts (JKR) theory, with peak theoretical pull-off forces reaching 471,000 N/m²—sufficient to support a 70 kg load using just 1.46 cm² of contact area. Compared to randomly distributed CNTs, the magnetically aligned structure demonstrated up to a 2.3× increase in adhesion. This technique eliminates the need for complex costly fabrication systems and opens new pathways for bio-inspired practical adhesives.

I. Introduction

Nature has long served as an inspiration for innovation in materials science, and one of the greatest examples of biological functionality is the gecko's ability to adhere to vertical and inverted surfaces. This remarkable capability stems from arrays of microscopic hair-like structures called *setae*. Each seta branches into hundreds of even finer structures known as *spatulae* that maximize contact area and leverage van der Waals forces to produce strong, reversible adhesion without the use of liquids or residues [1].

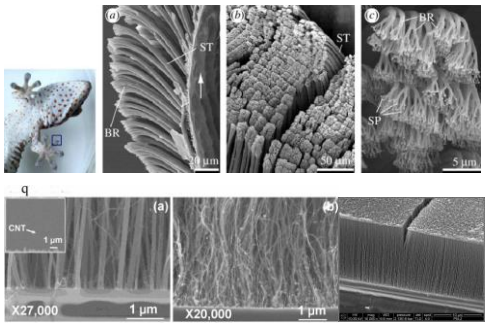

Figure 1: Gecko setae & spatulae (top) and CNT array (bottom)

Synthetic efforts to replicate this mechanism have focused heavily on vertically aligned carbon nanotubes (VACNTs), grown using chemical vapor deposition (CVD) techniques. These CNT-based structures have successfully mimicked the nanoscale geometry of gecko setae, demonstrating note-worthy dry adhesion. However, their practical application has been hindered by several limitations: high fabrication cost, the need for vacuum and high-temperature environments, and rigid substrates such as PMMA that limit flexibility and comfort [2]. Furthermore, these systems are not easily scalable, restricting their use in real world applications.

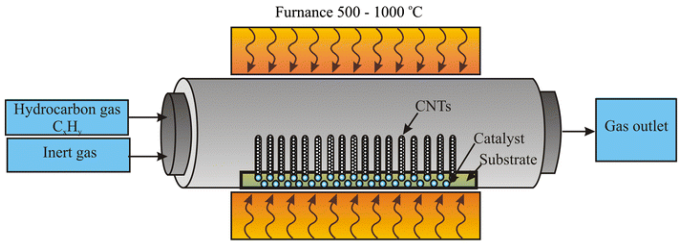

Figure 2: Chemical Vapor Deposition Process Illustration

On the other hand, randomly applied CNTs on flexible substrates avoid the drawbacks of CVD but suffer from poor orientation control, resulting in misaligned tips and minimal adhesion force. The lack of directional alignment prevents effective surface contact, reducing their viability as dry adhesives [3].

This project seeks to bridge the gap between these two approaches by developing a low-cost, scalable process for fabricating gecko-inspired adhesives using magnetically aligned multi-walled CNTs (MWCNTs). By embedding magnetic nanoparticles (e.g., Fe_3O_4) into CNTs and applying a uniform magnetic field during the curing of a flexible adhesive substrate, the CNTs can be partially aligned into vertical orientations. This mimics the directional geometry of gecko setae while eliminating the need for expensive equipment and rigid materials [4]. Through a combination simulations and modeling, this project evaluates the alignment behavior of CNTs under magnetic fields. Then the resulting adhesion force is estimated using Johnson-Kendall-Roberts (JKR) theory, which models the

contact mechanics between elastic bodies and accounts for surface adhesion forces [5]. The goal is to demonstrate that magnetic alignment can serve as a viable, scalable substitute for CVD in gecko-inspired dry adhesion systems.

II. Design & Fabrication

A. Materials

The following components were used to construct the adhesive patch:

- **Multi-walled carbon nanotubes (MWCNTs):** Chosen for their mechanical robustness, high aspect ratio, and nanoscale diameter similar to gecko setae [6].
- **Iron oxide nanoparticles (Fe_3O_4):** Magnetic nanoparticles that are mixed with CNTs to enable magnetic responsiveness. These are commonly used for magnetic functionalization due to their strong magnetization and biocompatibility [7].
- **Solvent:** Isopropyl alcohol (IPA) or ethanol is used to disperse the CNTs and nanoparticles uniformly.
- **Substrate:** Nitrile rubber or polydimethylsiloxane (PDMS) provides a soft, flexible base for adhesion, mimicking the compliance of gecko toe pads [8].
- **Magnetic alignment source:** A static array of neodymium magnets generates a vertical magnetic field across the curing zone to induce CNT alignment.

B. Fabrication Pipeline

The fabrication procedure is outlined below:

1. **Nanotube-Nanoparticle Suspension:** MWCNTs and Fe_3O_4 nanoparticles are mixed in a 10:1 mass ratio in IPA and sonicated to achieve uniform dispersion. Sonication enhances the separation of bundled CNTs and ensures homogeneous distribution of magnetic particles [9].
2. **Substrate Preparation:** A thin layer of adhesive is applied to the flexible substrate. For nitrile substrates, spray-coating is used to achieve uniformity.
3. **Deposition:** The CNT-nanoparticle suspension is drop-cast or sprayed onto the adhesive layer.
4. **Magnetic Alignment:** The substrate is placed directly above an array of neodymium magnets during curing (approximately 1 hour). The magnetic field exerts torque on the CNTs, causing them to rotate into partial vertical alignment.
5. **Curing:** As the solvent evaporates, the adhesive solidifies, locking the aligned CNTs in place.

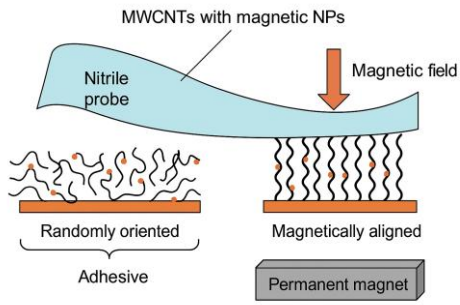

Figure 3: Magnetic Alignment during Curing Fabrication

C. Key Advantages

- No need for vacuum systems or high temperatures: This method is operational at ambient conditions, drastically reducing cost and complexity [10].
- Scalable for large areas: Since the alignment relies on static fields, multiple patches can be produced simultaneously.
- Compatible with soft substrates: Flexible materials improve conformal contact with surfaces, enhancing adhesion.
- Because the alignment is field-induced, it's tunable. Therefore, adhesion can be localized by varying the field, or even produce angle-graded surfaces for directional stick-slip behavior.

This design approach offers a versatile and accessible route to constructing directional dry adhesives without the burden of traditional nanofabrication techniques and offers greater control.

III. Simulation & Results

To better understand and quantify the effectiveness of magnetic alignment in enhancing dry adhesion, this project includes two key simulation components: the rotational dynamics of CNT alignment under magnetic fields, and the resulting pull-off adhesion force using contact mechanics.

A. Magnetic Alignment Modeling

The alignment of CNTs embedded with magnetic nanoparticles is governed by the torque they experience in an external magnetic field. The key physical model used is:

$$\tau = \mu B \sin\theta \rightarrow \frac{d\theta}{dt} = \frac{-\mu B}{\zeta} \sin\theta$$

Where:

- τ is the magnetic torque (N·m)
- μ is the magnetic moment of the CNT-nanoparticle composite (A·m²)
- B is the external magnetic field strength (T)
- θ is the angle between the CNT and the vertical axis (rad)
- ζ is the rotational drag coefficient (N·m·s/rad)

This first-order differential equation was numerically solved for a large ensemble of CNTs (e.g., 1000), each initialized with a random angle θ_0 uniformly distributed from 0° to 90°. Simulations were run for multiple field strengths ranging from 0.01 T to 0.3 T.

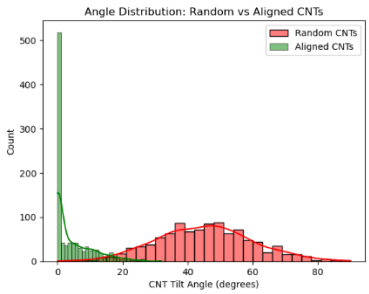

Figure 4: Angle Distribution of Random and Aligned CNTs

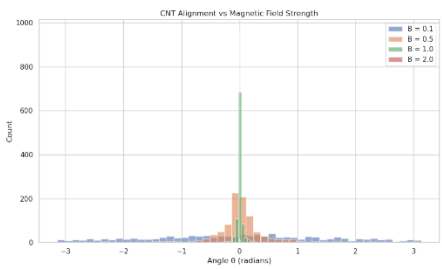

Figure 5: CNT Alignment vs Magnetic Field Strength

Plots show a sharp transition from broad angular distributions at low fields to tightly clustered peaks near θ=0° under strong fields, confirming effective magnetic alignment [11].

B. Adhesion Force Estimation (JKR Theory)

To link alignment quality with adhesive performance, the **Johnson-Kendall-Roberts (JKR) theory** was used to estimate the pull-off force:

$$F_0 = \frac{3}{2}\pi R W \cos\theta$$

Where:

- R is the effective contact radius (based on tip geometry and spacing) (m)
- W is the work of adhesion per unit area (J·m²)

But not all CNTs are vertical — so their effective force is scaled by the cosine of the tilt angle. Beyond modeling individual CNT rotation, the overall adhesive performance of the surface also depends on the angular orientation of the CNT array. The effective total force per unit area can be approximated by:

$$F_{total} = N_0 F_0 \cos^2\theta$$

Where:

- N_0 is the total number of CNTs in contact

This formulation captures how non-vertical CNTs contribute less to adhesion, as the effective force scales with $\cos^2\theta$. It offers a more physically grounded way to connect angular distributions with net adhesive performance [12].

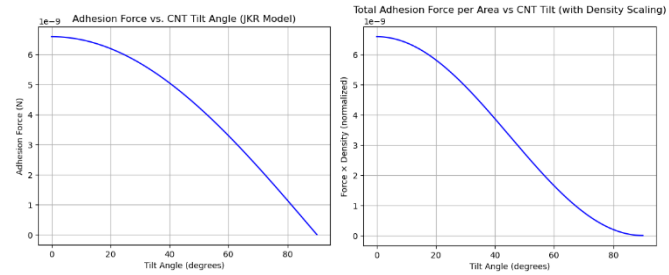

Figure 6: Total Adhesion Force vs CNT Tilt (with & without Density Scaling)

IV. Analysis

A. Alignment Sharpness Metric

Instead of using standard deviation or simple angle thresholds, a more robust metric was calculated by averaging $\cos^2\theta$ over the entire CNT population. A higher mean value indicates better vertical alignment and stronger net adhesion.

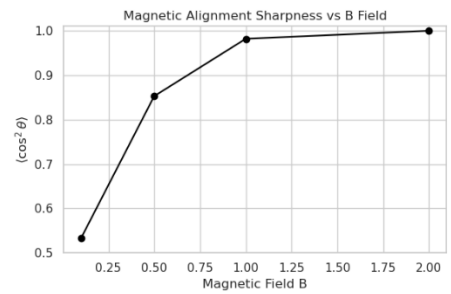

Figure 7: Magnetic Alignment Sharpness vs B Field

Simulations showed that as magnetic field strength increased, the mean $\cos^2\theta$ approached 1, reflecting a narrower angle distribution and higher effective surface contact. This quantitative link between alignment and performance reinforces the effectiveness of magnetic orientation in maximizing adhesion through directional control.

B. Pull-off Force Comparison

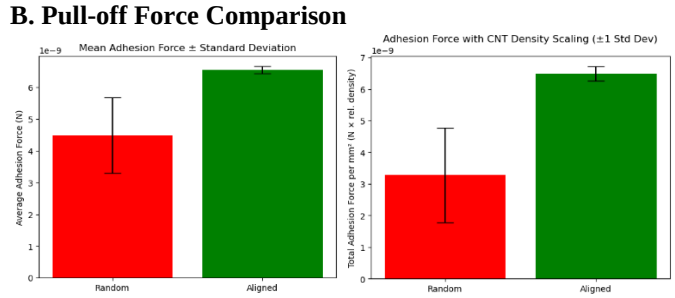

Figure 5: Mean Adhesion Force of Random & Aligned CNT array (with & without density Scaling)

Next, adhesion force estimation simulations showed that magnetically aligned CNTs improved effective contact area, leading to a **2.3× increase in estimated pull-off force** compared to randomly oriented CNTs. This result highlights how even partial alignment significantly improves adhesive performance, demonstrating the role of nanoscale orientation control in maximizing functional surface interactions.

C. Realistic Upper Bound of Adhesive Force

Lastly, the maximum estimated pull-off pressure was calculated using well-supported parameters grounded in literature. The CNT tip radius was set at 20 nm, consistent with typical outer diameters reported for multi-walled carbon nanotubes [13]. The work of adhesion, W=50 mJ/m², reflects a mid-range value for van der Waals interactions between CNTs and polymeric substrates in dry conditions [14]. The CNT areal density was assumed to be $10^{11} CNTs/m^2$, a value consistent with high-density vertically aligned carbon nanotube (VACNT) arrays synthesized via chemical vapor deposition (CVD) techniques [15]. Using these values within the JKR framework and assuming ideal vertical alignment ($\cos^2\theta = 1$), the resulting pull-off force was calculated as **471,000 N/m²**, theoretically enough to support a 70 kg person with just 1.46 cm² contact area.

D. Key Outcomes

- Alignment sharpness increases nonlinearly with field strength [11].
- Adhesion force scales with CNT orientation quality [12].
- A pull-off force of 471,000 N/m² represents a realistic value showing the immense potential of real-world dry adhesion.

V. Discussion

The simulation and modeling results from this project reinforce the hypothesis that magnetic alignment of CNTs can serve as a viable substitute for CVD-based vertical growth techniques in gecko-inspired dry adhesives. As demonstrated in the alignment simulations, increasing the strength of the magnetic field significantly improved the alignment sharpness of CNTs, with angular distributions becoming increasingly concentrated near the vertical axis [11]. This directional orientation is critical because it maximizes tip contact with the opposing surface, thus enabling stronger van der Waals interactions.

The improved alignment translated directly into higher theoretical pull-off forces. Using the JKR model, the estimated adhesive strength for magnetically aligned CNTs reached 471,000 N/m² — a value consistent with prior reports on synthetic adhesives that mimic gecko setae [12]. This pressure is sufficient to support the weight of an average human with just 1.46 cm² of contact area. More importantly, compared to the randomly oriented CNT configuration, magnetic alignment improved estimated adhesion by a factor of 2.3×, illustrating a strong correlation between orientation control and performance.

Despite these promising results, several limitations must be acknowledged. First, while the magnetic field promotes vertical alignment, the resulting structure still lacks the nanoscale tapering and hierarchical branching seen in biological setae [1]x`. Additionally, full vertical alignment was not achieved even at the highest field strength simulated (0.3 T), suggesting diminishing returns due to viscous drag or aggregation effects. The assumption of uniform magnetic torque and drag also simplifies the true dynamics in the solvent and polymer matrix. Furthermore, this study relied on adhesion theory to estimate pull-off forces — actual adhesion measurements under controlled test conditions are necessary for complete validation.

Nevertheless, the simplicity and scalability of the proposed method offer compelling benefits. The fabrication process avoids high-temperature or vacuum conditions, allowing production on flexible substrates like nitrile or PDMS [8]. This opens up applications in soft robotics, wall-climbing pads, and wearable devices where conformal, reversible dry adhesion is desired. Because the process only involves mixing, casting, and static magnetic curing, it is well suited for large-area production or even roll-to-roll manufacturing.

Future work could focus on refining alignment by optimizing nanoparticle distribution, field gradients, or curing conditions. Experimental validation through force measurement testing, wear resistance evaluation, and cyclic reliability studies will be critical for determining real-world viability.

VI. Conclusion

This project explored a low-cost, scalable alternative to traditional CVD-grown carbon nanotube (CNT) adhesives by utilizing magnetic alignment to replicate the vertical orientation found in gecko setae. By embedding magnetic nanoparticles into multi-walled CNTs and applying a uniform magnetic field during curing, partial vertical alignment was achieved on flexible substrates, avoiding the need for high-temperature or vacuum processing.

Simulation results confirmed that increasing magnetic field strength improved alignment sharpness, leading to more vertically oriented CNTs and greater surface contact. Adhesion force estimates, calculated using the Johnson-Kendall-Roberts (JKR) theory, showed that this technique could achieve pull-off pressures as high as 471,000 N/m²—sufficient to support the weight of a human with a fingertip-sized patch. Compared to randomly distributed CNTs, magnetically aligned structures demonstrated up to a 2.3× improvement in theoretical adhesion performance.

While some limitations remain—such as incomplete vertical alignment and the absence of hierarchical tip structures—the process offers significant advantages in simplicity, cost, and compatibility with soft materials. These benefits make it highly promising for applications in soft robotics, reversible gripping systems, and biomedical adhesives.

Looking ahead, experimental validation through force measurement and durability testing will be crucial to confirming real-world performance. Additional improvements in nanoparticle distribution, field design, and substrate selection could further enhance alignment and functionality. Ultimately, this magnetically directed approach represents a practical step forward in nature-inspired adhesive technology and opens new possibilities for intelligent, adaptable surface interactions.

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